Reg. No. | | | | | | | | | | | [12ME71]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER-2015

MECHANICAL VIBRATIONS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. Define Resonance. **02**

   b. A harmonic motion is given the equation $x(t)=5\sin(30t+70^\circ)$cm. where t is in seconds. Find i. period of motion, ii. Maximum displacement iii. velocity, & iv. Acceleration. **04**

   c. A body is subjected to harmonics motions $X_1=10\sin(3t+60^\circ)$ and $x_2=5\text{Cos}(3t+30^\circ)$. What harmonic motion should be given to the body to bring it to equilibrium? **08**

2. a. Determine the differential equation and the natural frequency of the system shown in the fig. 2(a) **07**

   b. Determine the natural frequency of the spring mass pulley system as shown in the fig.2.(b). **07**

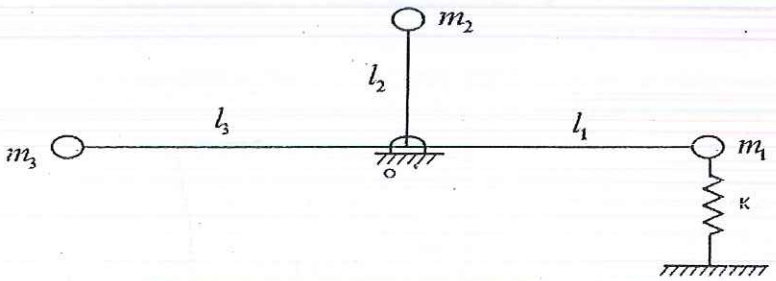

Fig. 2.(a)

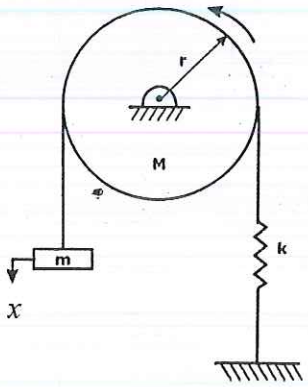

Fig. 2.(b)

3. a. Derive an expression for logarithmic decrement **07**

   b. A shock absorber is to be designed so that its over shoot is 10% of the initial displacement when released. Determine the damping factor. If the daming factor is halved in this value what will be the over shoot? **07**

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4. a. A mass of 100 kg been mounted on a spring dashpot system having spring stiffness of 2000kg/m and damping co-efficient of 120 N-sec/m. The mass is acted upon by a harmonic force of 40N of the un damped natural frequency of the system determine **07**

i. Amplitude of vibration of the moss

ii. Force transmissibility ratio.

b. A machine unit having a mass of 40 kg is to be supported on three springs. Each having a stiffing K. unit operates at 500 rpm . Find the stiffness value K if only 10% of the shaking force is allowed to be transmitted to the supporting structure. **07**

5. a. Define whirling of shaft. **02**

b. Explain briefly any one of the commonly used seismic transducer in vibration measurement. **06**

c. A rotor of mass 12 kg is mounted midway on a 25mm diameter horizontal shott supported at the ends at two bearings. The span between the bearing is 0.9 m. find the its critical speed if $E=200x\ 10^9\ N/m^2$. **06**

6. Determine **14**

i. Equation of motion

ii. frequency equation and natural frequency of the system, and

iii. nodal vectors and mode shapes. For the system in fig 6.

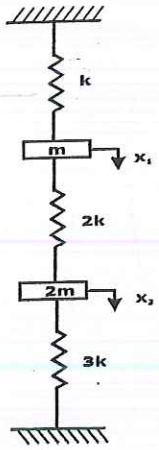

Fig. 6.

7. a. Find the natural frequency of the system shown in fig.7.(a) by Dunkerley's method. E=200Gpa, $I=5 \times 10^{-7} m^4$. **06**

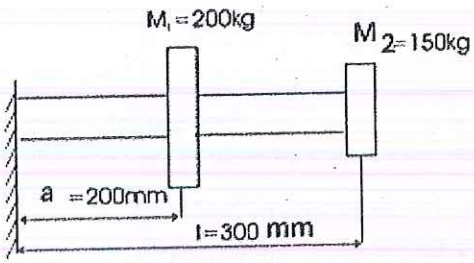

Fig-7(a)

b. Calculate all the influence co-efficient of the system shown in fig.7(b) **08**

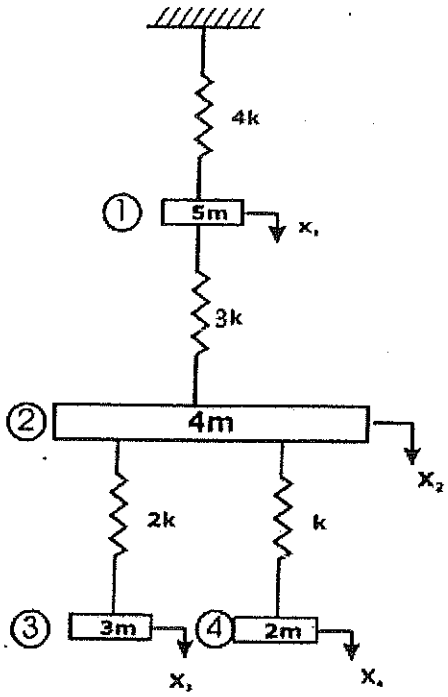

Fig-7(b)

8. Show that torsional vibration of rod **14**

$$\Theta(x, t) = \delta\left[A_n \cos\left(\frac{\varpi_n}{c}\right)x + B_n \sin\left(\frac{\varpi_n}{c}\right)x\right][C_n Cos\varpi_n t + D_n Sin\varpi_n t]$$

Where 'ϖ' is the frequency of vibration & A_n, B_n, C_n, D_n are arbitrary constants.